

NPTEL Course on "Computer System Optimization for ML"

Lecture #43

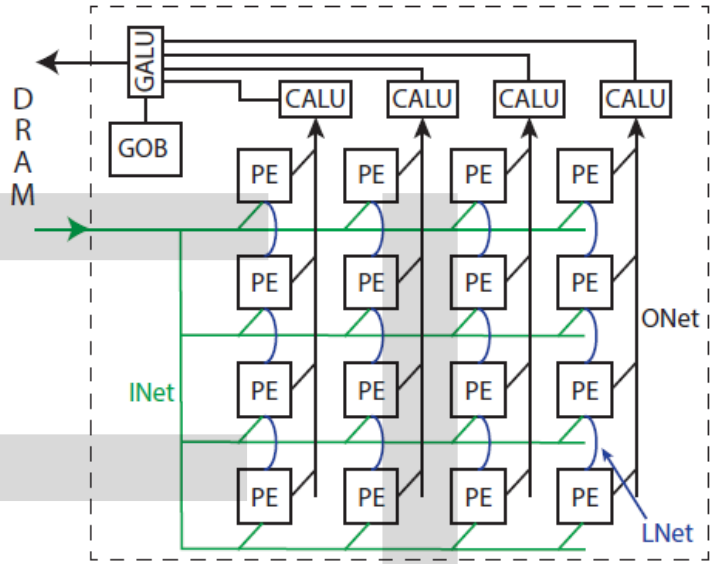
Agenda

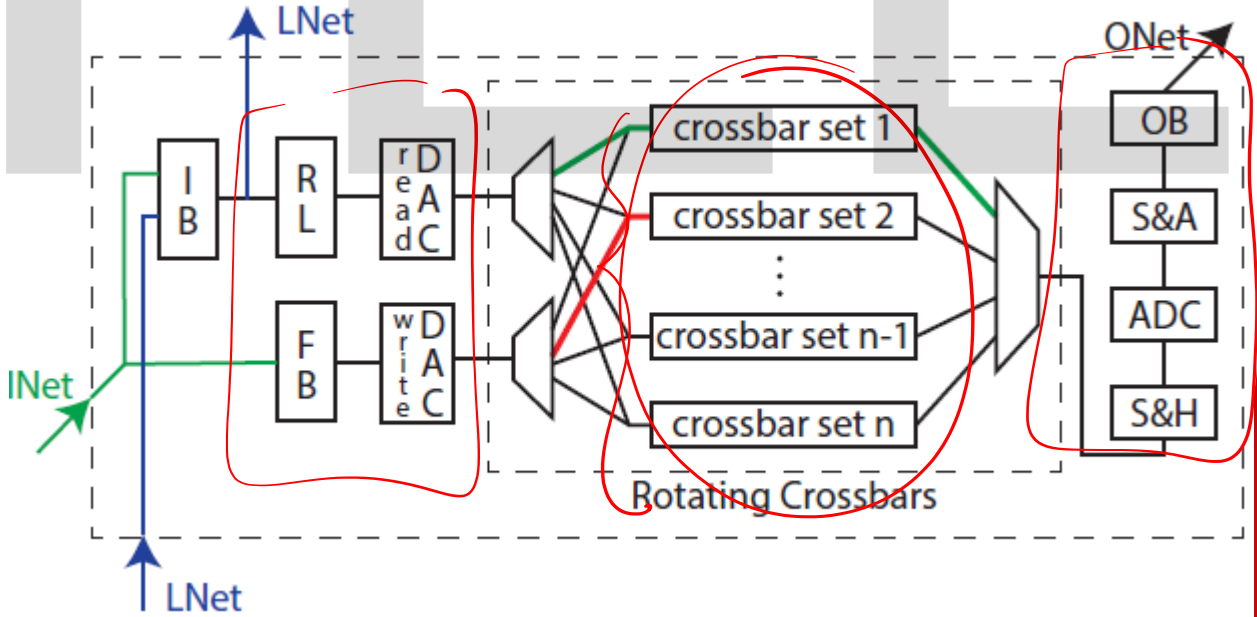
- Recap
- Simulation Platform
- Area-aware Optimization
- Communication-aware Optimization

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AtomLayer Architecture

- **Processing elements (PE), on-chip network, ALU tree, global output buffer**
- **ALU tree**
  - Column ALU (CALU)
  - Global ALU (GALU)
- **PE**
  - Rotating crossbars
  - Peripheral devices
  - Four buffers





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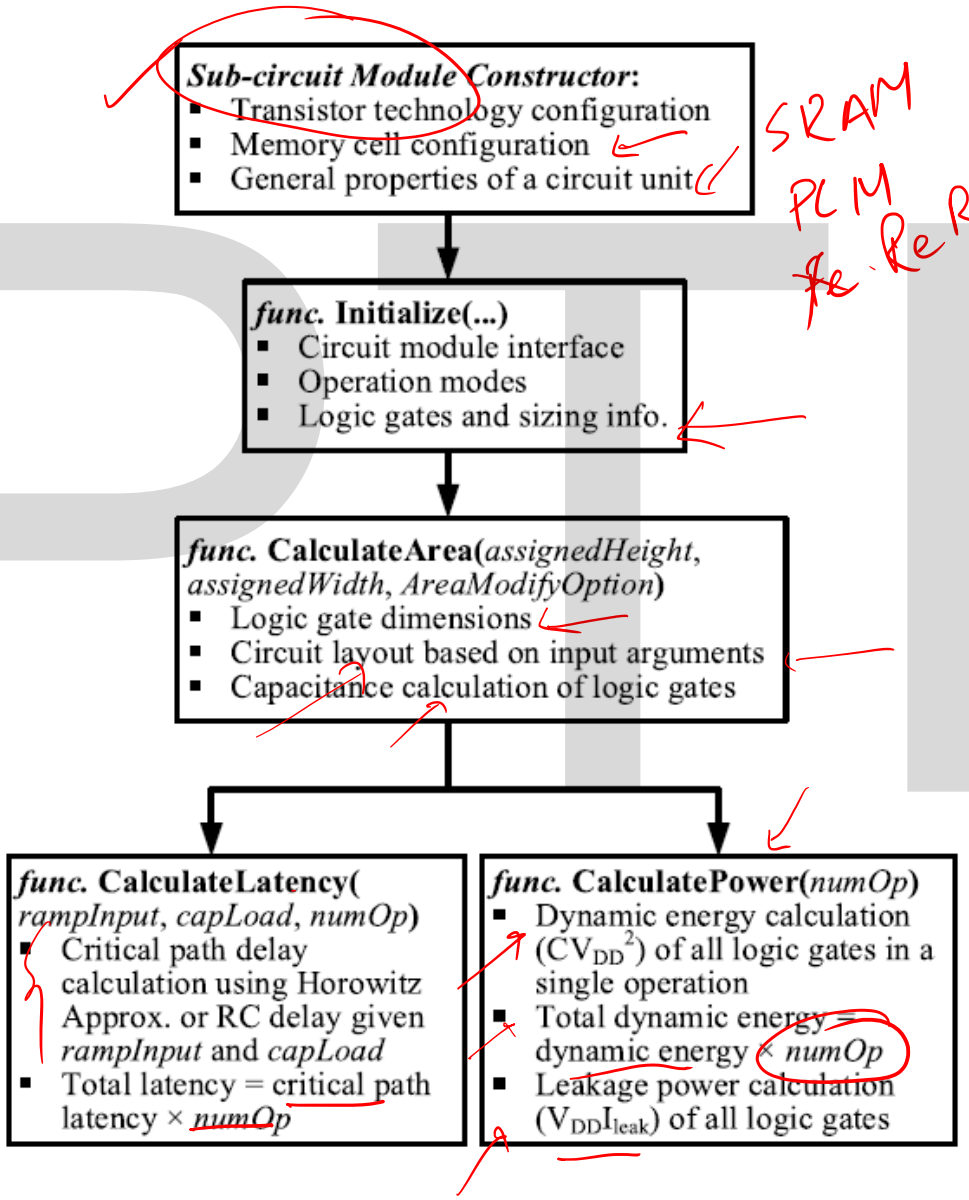
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NeuroSim: A Circuit-Level Macro Model for Benchmarking Neuro-Inspired Architectures in Online Learning

- **A performance analysis tool for in-memory computing based system**
- **The architecture is similar to ISAAC**
  - Tile-based
  - Tiles are connected with on-chip interconnect
  - Tiles consist of crossbar-based IMC elements
- **Analytical performance models**
  - Buffers
  - ADC
  - IMC elements
- **Estimates area, energy, latency, throughput**

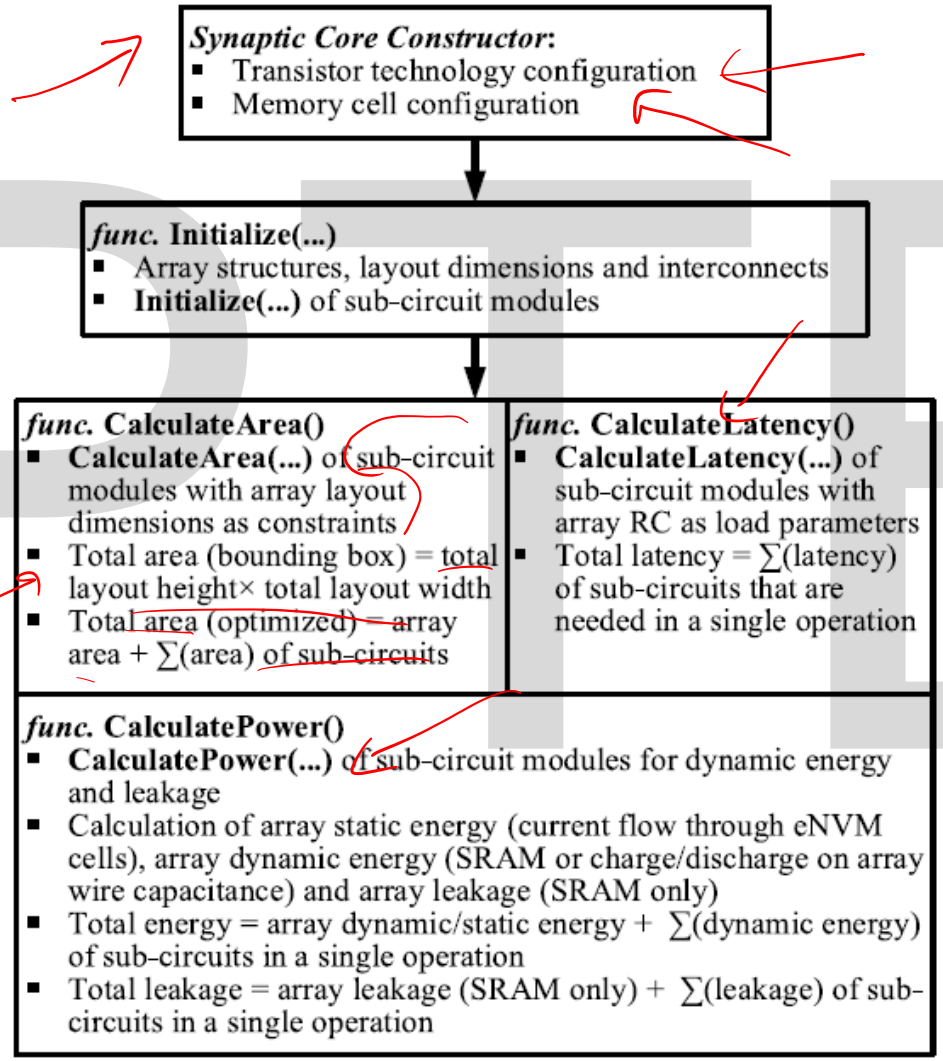
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Circuit-level Execution Flow



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Architecture-level Execution Flow



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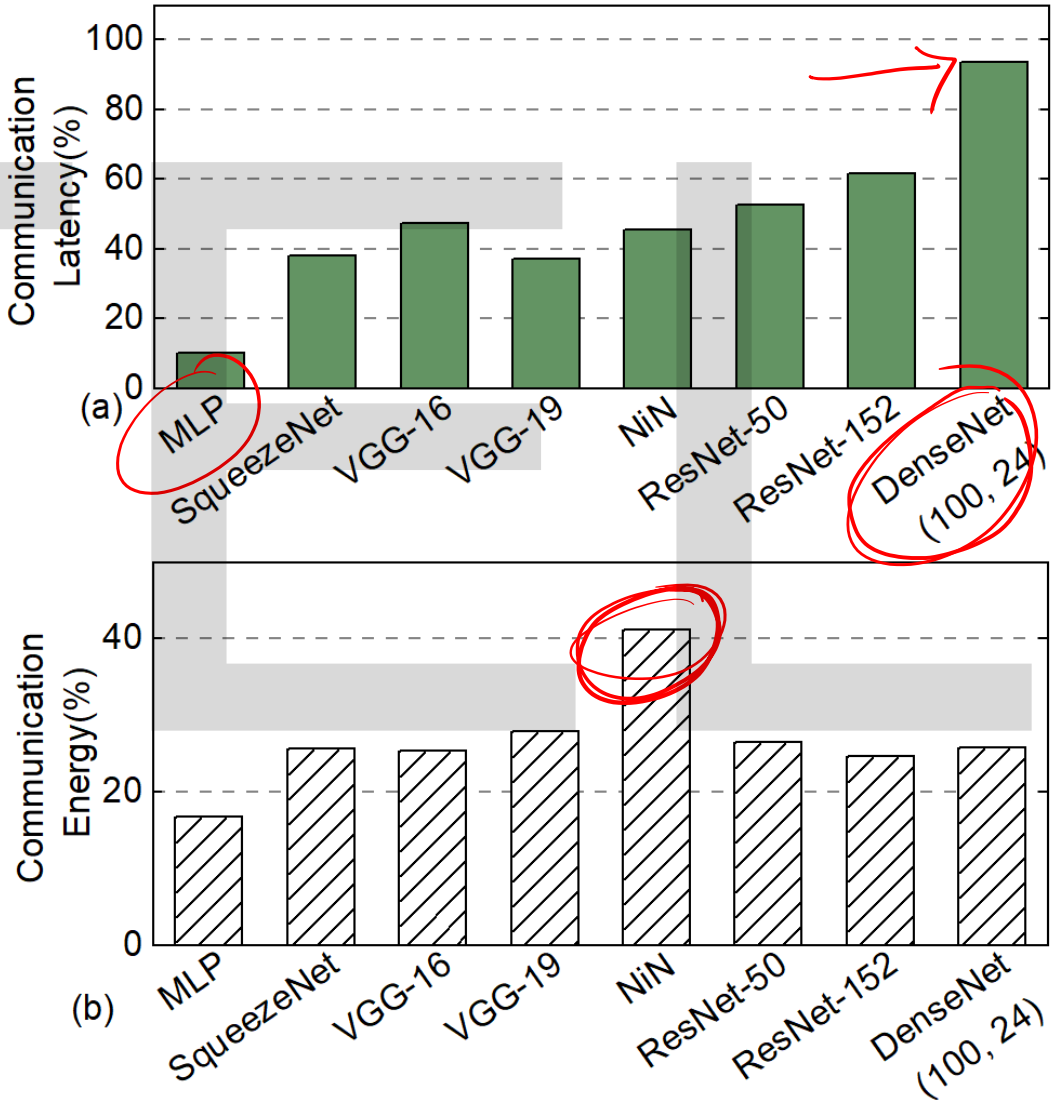
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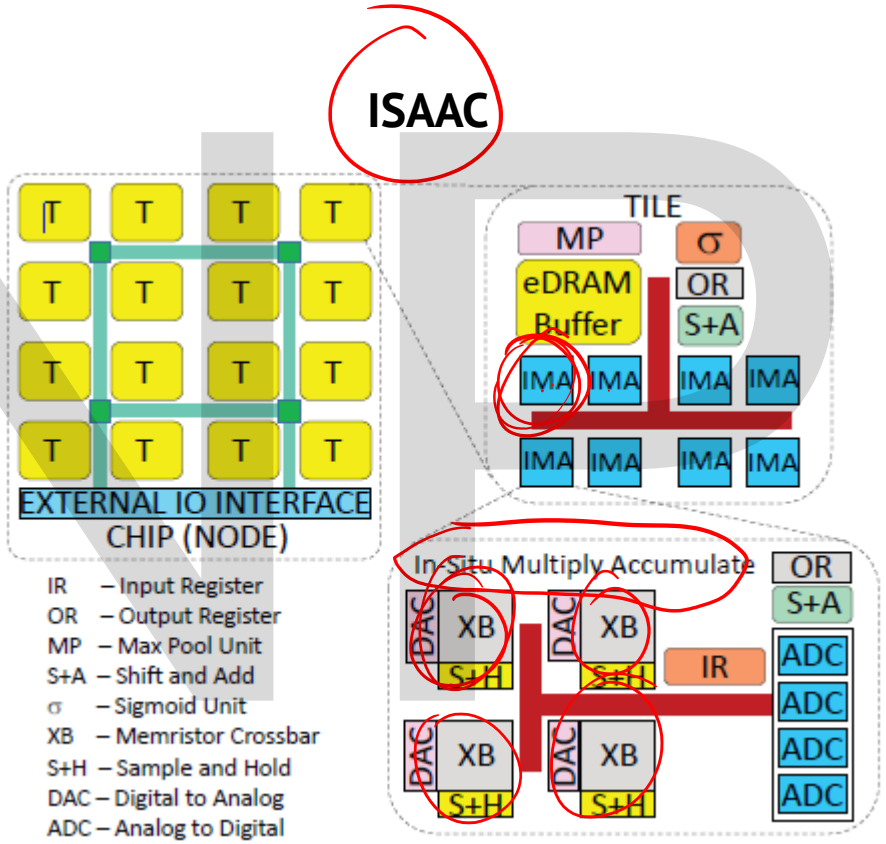
Bottleneck of the DNN Accelerator: Communication

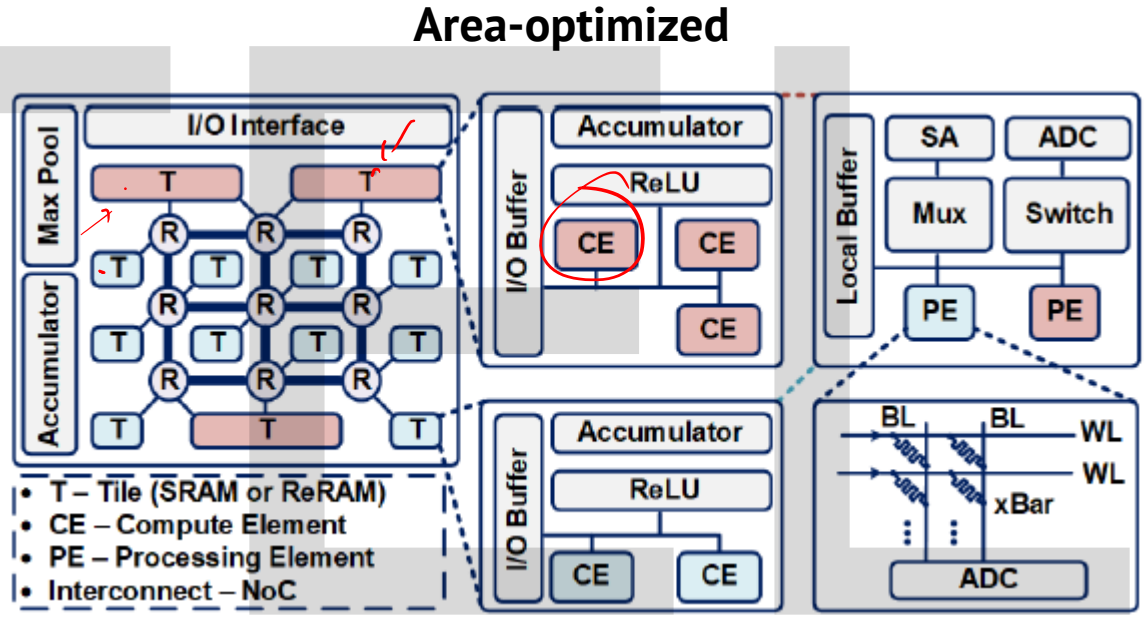
- Communication volume increases with deeper and denser DNNs
- Target hardware: In-memory Computing based
  - Exhibits less latency from off-chip memory accesses
- Up to 90% of the inference latency comes from communication
- Up to 40% of the energy is consumed by communication
- Need for an efficient on-chip communication strategy for DNN accelerators



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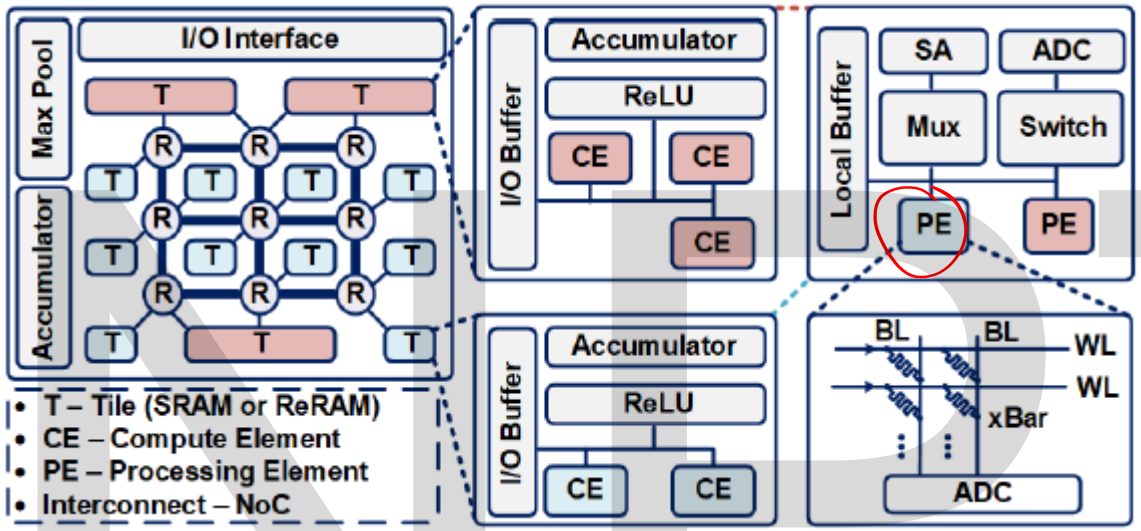
Area-aware Optimization [1]





[1] Krishnan, Gokul, et al. "Interconnect-aware area and energy optimization for in-memory acceleration of DNNs." *IEEE Design & Test* 37.6 (2020): 79-87.

Reducing Residual Area



Symbol	Definition	Symbol	Definition
T_k	Number of tiles in k^{th} layer	N_{bits}	Weight precision
Kx_k, Ky_k	Kernel size	c_k	Number of CEs in T_k
$(PE_x)_k$, $(PE_y)_k$	Size of the PE array	p_k	Number of PEs in c_k
I_k	Input activations in k^{th} layer	N_k^{if}, N_k^{of}	Number of i/p and o/p features

$$N_k^r = \left\lceil \frac{Kx_k \times Ky_k \times N_k^{if}}{(PE_x)_k} \right\rceil$$ → Required number of rows of PEs

$$N_k^c = \left\lceil \frac{N_k^{of} \times N_{bits}}{(PE_y)_k} \right\rceil$$ → Required number of columns of PEs

$$T_k = \left\lceil \frac{N_k^r \times N_k^c}{c_k \times p_k} \right\rceil$$

$$(c_k \times p_k \times T_k - N_k^r \times N_k^c)$$ → Residual Area

$$A_k(c_k, p_k) = (c_k \times p_k \times T_k - N_k^r \times N_k^c) \times T_k^2$$ → Cost function

$$\underset{c_k, p_k}{\text{minimize}} \quad A_k(c_k, p_k); \; k = 1, \ldots, K,$$

$$\text{subject to} \quad c_{min} \le c_k \le c_{max}, \quad p_{min} \le p_k \le p_{max}.$$

→ Constrained optimization

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THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)